

ASSIGNMENT GUIDELINES

- Make the changes in the PPT as you solve the parts
- This file contains the template for the **EDA part** of the project.
- Check the instructions added in the note section of every slide for clarity.
- Don't move around any image or text box
- If you require more/lesser elements, be careful when you copy/delete the existing ones.

ASSIGNMENT

Name: Tania Sharma

Problem Statement

Bank of Corporate conducted a telemarketing campaign for one of its financial products, 'Term deposits', to build a long-term relationship with the existing customers. Your goal is to identify the target customers for the term deposits from the pool of the bank's existing customers. You should also capture the key driving factors (or driver variables) behind the successful conversion of a customer, i.e., the customer opening a term deposit account with the bank.

Assignment Objective

To identify the target customers and the driving factors behind a successful customer conversion for future marketing campaigns.

PART I : Univariate Analysis

Variable under consideration: Subscription

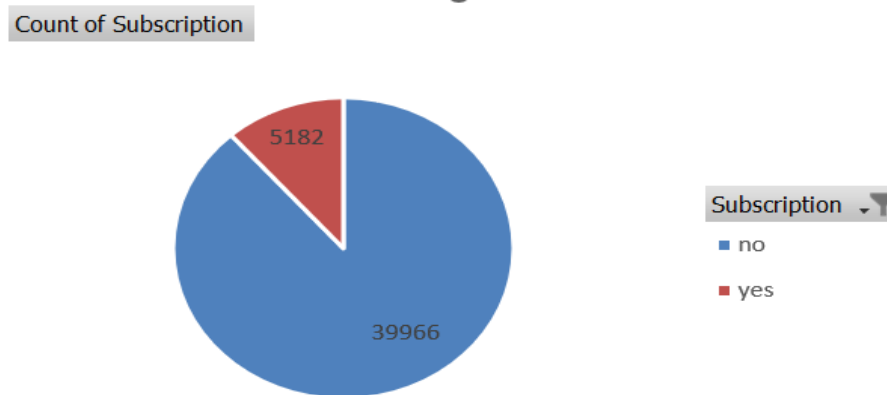
Results of Analysis:

- Total records analysed: 45,616
- Customers subscribed: 5,182
- Customers not subscribed: 39,966
- Subscription rate is comparatively low.

Insight: The majority of customers did not subscribe to the term deposit, indicating that future campaigns should focus on high-potential customer segments.

Chart:

Pie Chart or Bar Chart of Subscription (Yes/No)



PART I : Univariate Analysis

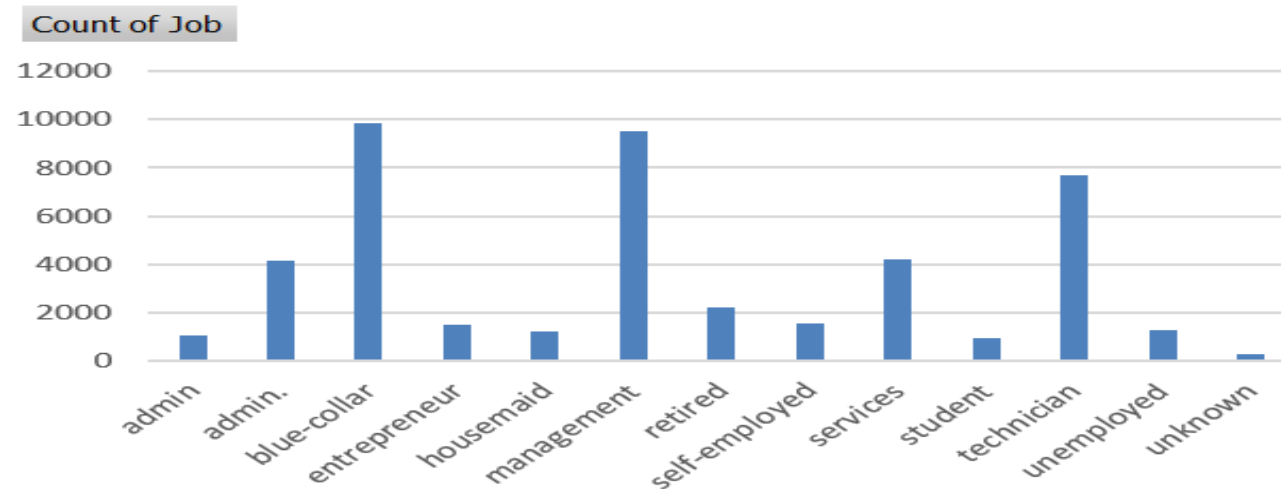
Variable under consideration: Job

Results of Analysis:

- Blue-collar workers form the largest customer group.
- Management and Technician categories also have significant representation.
- Students and retired customers form a smaller proportion of customers.

Insight : The customer base is dominated by working professionals and blue-collar employees.

Chart: Bar Chart of Job Distribution



PART I : Univariate Analysis

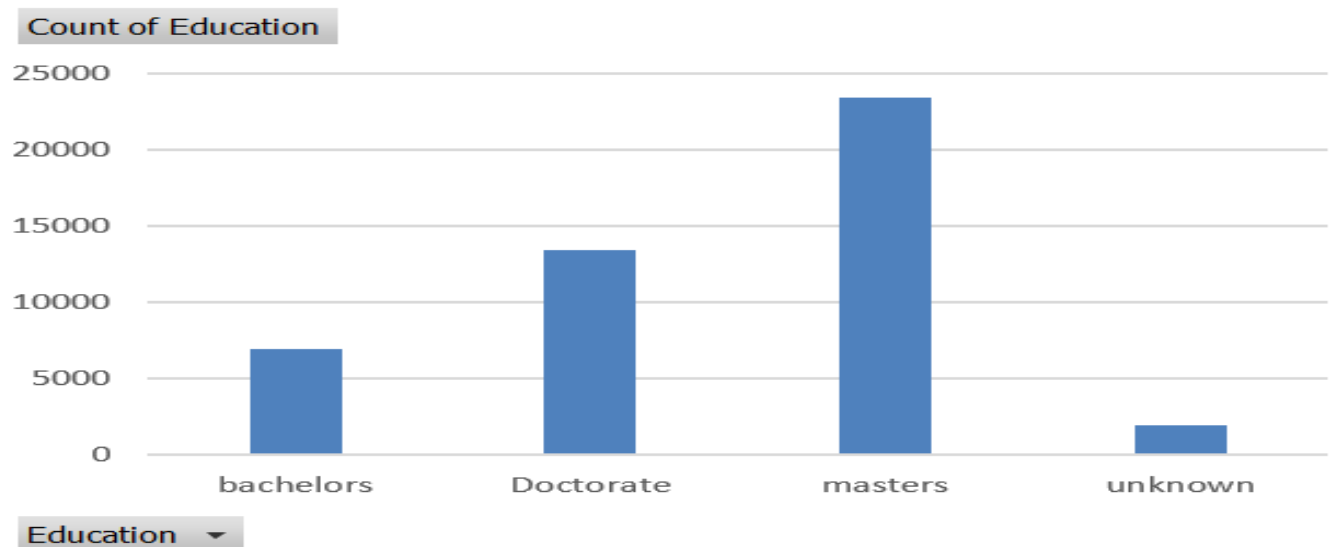
Variable under consideration: Education

Results of Analysis:

- Masters degree holders form the largest group.
- Doctorate holders represent a significant proportion.
- Bachelor degree holders form a smaller segment.

Insight : Most customers possess higher education qualifications, suggesting good awareness of banking products.

Chart : Bar Chart of Education Distribution



PART I : Univariate Analysis

Variable under consideration:Housing Loan

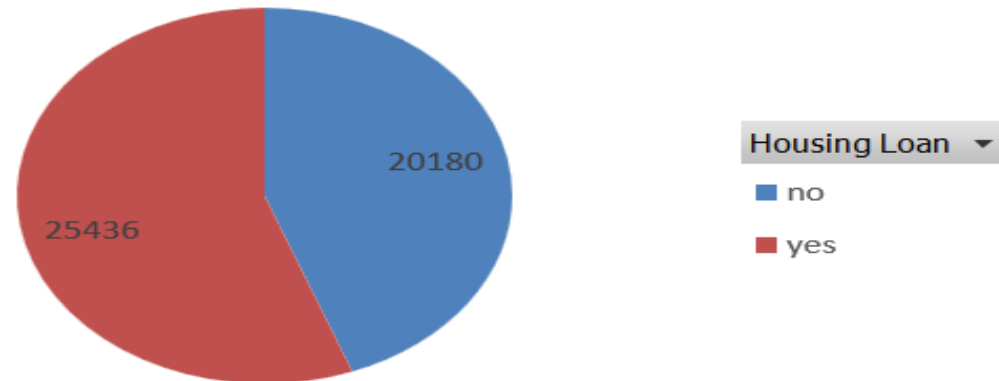
Results of Analysis:

- More than half of the customers have housing loans.
- Customers without housing loans represent a smaller segment.

Insight: Housing loans are common among existing customers and may influence investment decisions.

Chart: Pie Chart of Housing Loan (Yes/No)

Count of Housing Loan



PART II : Bivariate Analysis

Variables under consideration: Housing Loan vs Subscription

Results of Analysis:

Housing Loan	Subscription Rate
No	16.42%
Yes	7.56%

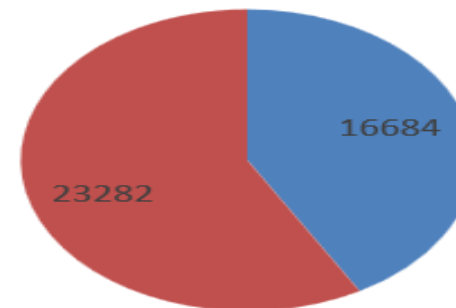
Insight: Customers without housing loans are more than twice as likely to subscribe to term deposits.

Business Interpretation: Customers with existing housing loan obligations may have lower disposable income available for investments.

Chart: Clustered Bar Chart

A significant proportion of customers have housing loans

Count of Subscription



Housing Loan ▼

- no
- yes

Subscription ▼

PART II : Bivariate Analysis

Variables under consideration: Personal Loan vs Subscription

Results of Analysis:

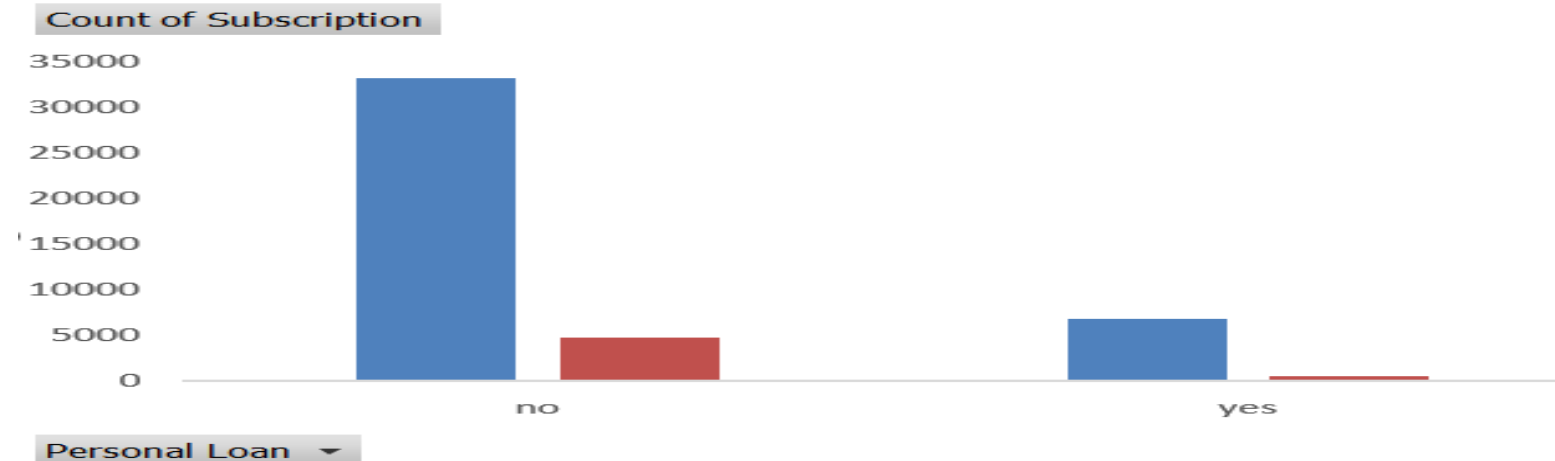
Personal Loan	Subscription Rate
No	12.42%
Yes	6.57%

Insight: Customers without personal loans show significantly higher subscription rates.

Business Interpretation: Debt-free customers appear more willing to invest in long-term deposit products.

Chart: Clustered Bar Chart

Customers without personal loans are more likely to subscribe.



PART II : Bivariate Analysis

Variables under consideration: Previous Campaign Outcome (outcome) vs Subscription

Results of Analysis:

Previous Outcome	Subscription Rate
Success	63.82%
Other	16.49%
Failure	12.44%
Unknown	9.08%

Insight: Customers who responded successfully to previous campaigns have the highest probability of subscribing again.

Business Interpretation: Previous campaign success is one of the strongest indicators of future conversion.

Customers with successful previous campaign outcomes show the highest conversion rates.

PART II : Bivariate Analysis

Chart : Stacked Bar Chart

Count of Subscription

outcome

unknown

success

other

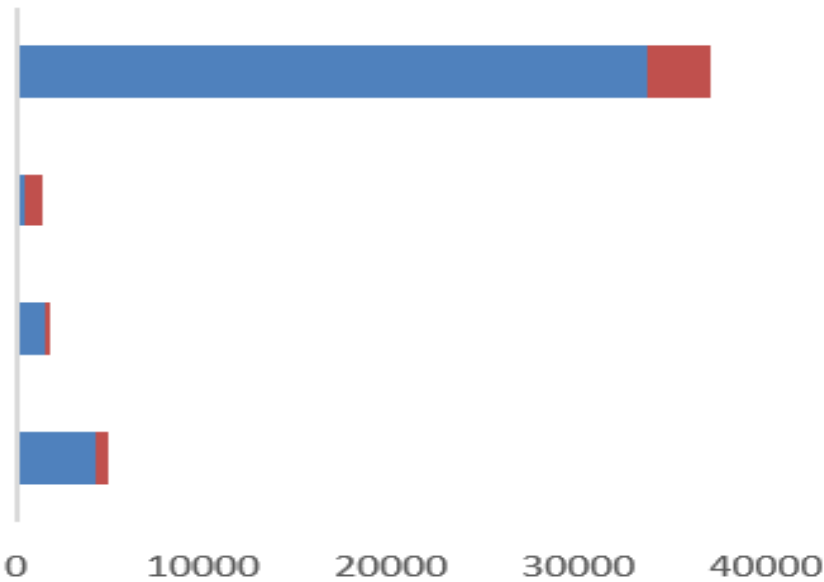
failure

Subscription

no

yes

0 10000 20000 30000 40000



PART II : Bivariate Analysis

Variables under consideration: Job vs Subscription

Results of Analysis:

Highest subscription rates:

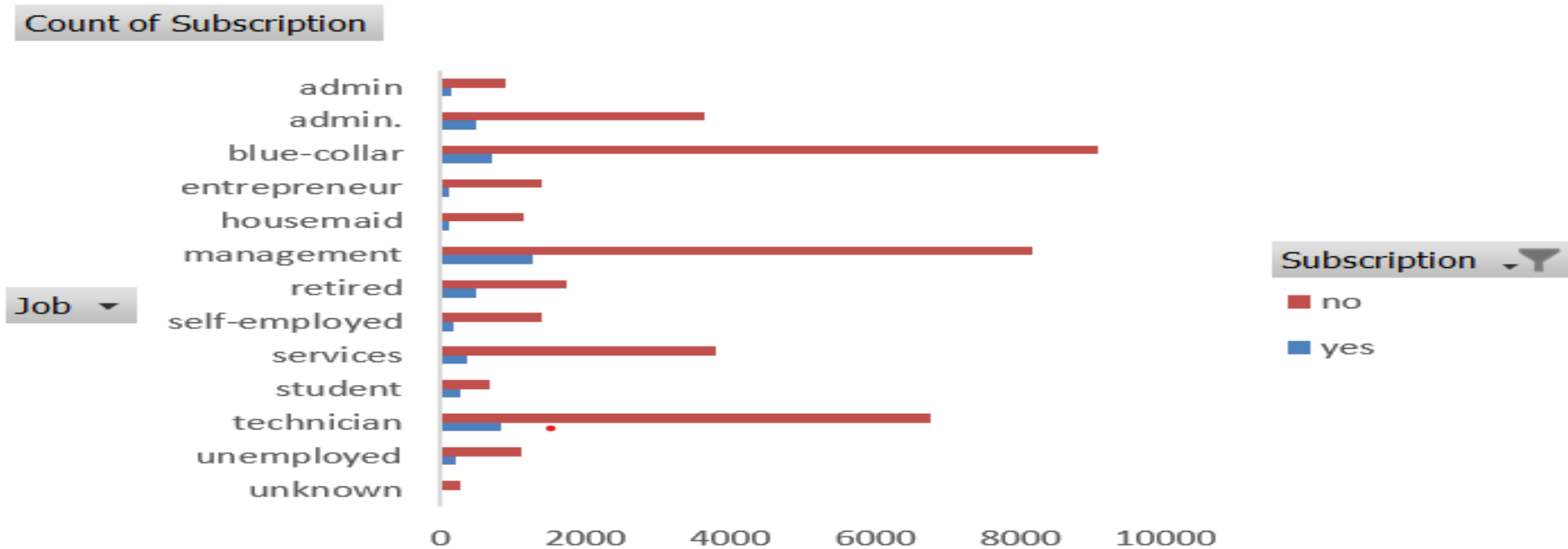
Job Category	Subscription Rate
Student	28.00%
Retired	22.43%
Unemployed	15.15%
Admin	13.83%
Management	13.52%

Insight: Students and retired customers are more likely to subscribe to term deposits.

Business Interpretation: These customer groups may prefer safer investment options with guaranteed returns.

PART II : Bivariate Analysis

Students and retired customers exhibit the highest subscription rates among occupational groups.



PART III: Major insights

Key Findings

Driver Variable 1: Previous Campaign Outcome

Customers who previously responded successfully show a subscription rate of 63.82%, making this the strongest predictor of conversion.

Driver Variable 2: Housing Loan Status

Customers without housing loans are significantly more likely to subscribe than customers with housing loans.

Driver Variable 3: Personal Loan Status

Customers without personal loans demonstrate greater interest in term deposits.

Driver Variable 4: Occupation

Students and retired customers have the highest conversion rates among all occupational groups.

Recommendations

- Prioritise customers who responded successfully in previous campaigns.
- Focus marketing efforts on customers without housing or personal loans.
- Target students, retired customers and management professionals.
- Design personalised campaigns for high-potential customer segments.
- Reduce marketing expenditure on low-response customer groups.

Conclusion

By focusing on previous campaign success, loan status and occupation, the bank can improve customer conversion rates and maximise the effectiveness of future term deposit campaigns.